

Quantum Entanglement, Part 3

Lecture 8: Metrics, Expansion, and a First Cosmology

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Chapter 1

Metrics, Expansion, and a First Cosmology

At the end of the preceding lecture we had reached a Lorentz-invariant wave equation and were preparing to write Maxwell's equations in spacetime form. An audience question now supplies a useful warning before we turn to gravitation: how much of a physical theory can symmetry determine by itself? The answer is emphatic: a great deal, but not everything. Lorentz invariance is a powerful filter on possible laws. It is not a substitute for the remaining physical assumptions or for experiment.

1.1 What symmetry does not determine

Question from the audience. Once the electric and magnetic fields are known to transform together under Lorentz transformations, do Maxwell's equations follow from relativity alone?

Response. No. Relativity constrains the form of the equations, but the construction also uses their linearity, the low order of their derivatives, the way the fields transform, and experimental information. Without those additional ingredients many Lorentz-covariant field equations would still be possible.

The most important of those extra assumptions for the opening discussion is linearity. If $A(x, t)$ is a solution of a linear field equation, then $\lambda A(x, t)$ is also a solution for every constant λ . If A and B are solutions, then so is

$$C(x, t) = A(x, t) + B(x, t). \quad (1.1)$$

This is the superposition principle. Two source-free electromagnetic waves may cross one another and emerge unchanged. They do not collide directly in the way material objects do. They can interact indirectly through charged matter, because each wave moves the charges and the moving charges radiate, but the vacuum field equations themselves are linear.

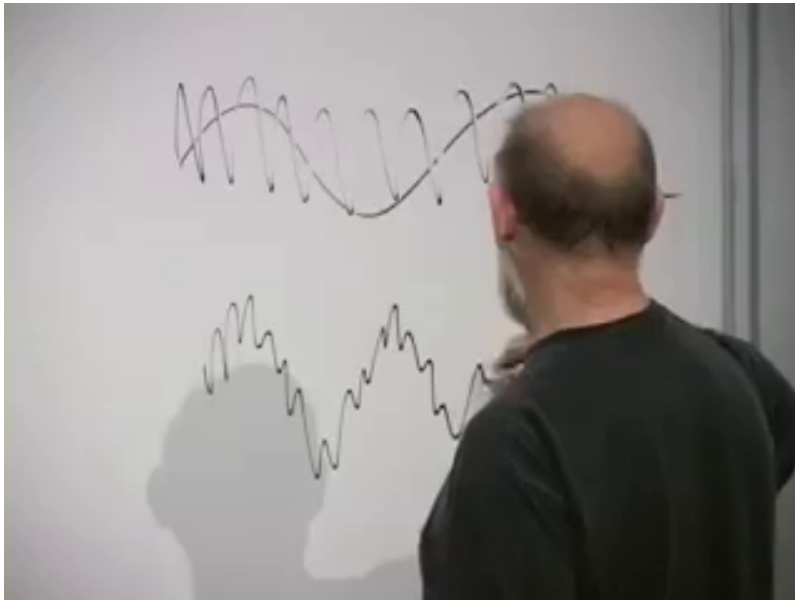


Figure 1.1: Two wave profiles on the board during the discussion of superposition. Their sum is again an allowed source-free wave.

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The harmonic oscillator makes the algebra transparent. Suppose

$$m\ddot{x} = -kx. \quad (1.2)$$

If x_1 and x_2 solve (1.2), then

$$m(\ddot{x}_1 + \ddot{x}_2) = -k(x_1 + x_2), \quad (1.3)$$

so $x_1 + x_2$ is a solution. Add a nonlinear term, however,

$$m\ddot{x} = -kx + px^2, \quad (1.4)$$

and the sum no longer works:

$$(x_1 + x_2)^2 = x_1^2 + x_2^2 + 2x_1x_2. \quad (1.5)$$

The cross term $2x_1x_2$ has no counterpart in the two separate equations. The two motions now influence one another.

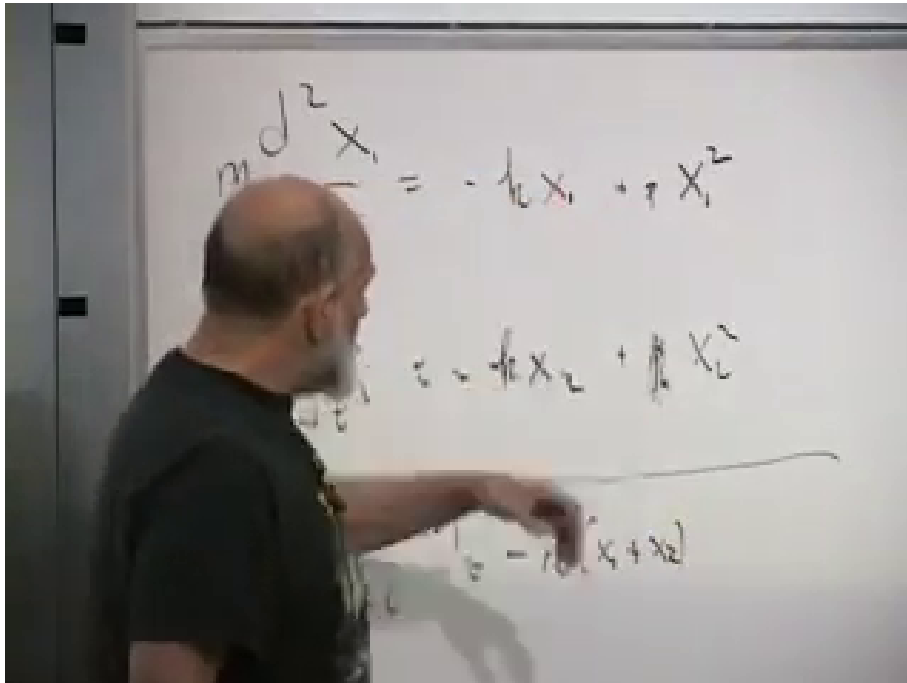


Figure 1.2: Separate oscillator equations compared with the equation for their sum. The quadratic term exposes the extra cross term that breaks superposition.

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This example fixes the methodological point that will remain in force throughout the cosmology discussion. Symmetry tells us which quantities may be combined and which equations are allowed. Dynamics and observation tell us which allowed equation nature uses.

1.2 The interval as geometry

We now leave electromagnetism and take a short route into general relativity and cosmology. The starting object is the interval between neighboring events. Label an event by four coordinates x^μ and a nearby event by $x^\mu + dx^\mu$. In inertial Cartesian coordinates, with $c = 1$ and the $(+, -, -, -)$ convention,

$$d\tau^2 = dt^2 - dx^2 - dy^2 - dz^2 = dx^\mu dx_\mu = \eta_{\mu\nu} dx^\mu dx^\nu. \quad (1.6)$$

The quantity $d\tau^2$, not any one coordinate difference, is invariant. Lorentz transformations change dt, dx, dy, dz while preserving this quadratic combination. Special relativity may therefore be read as the geometry of Minkowski spacetime.

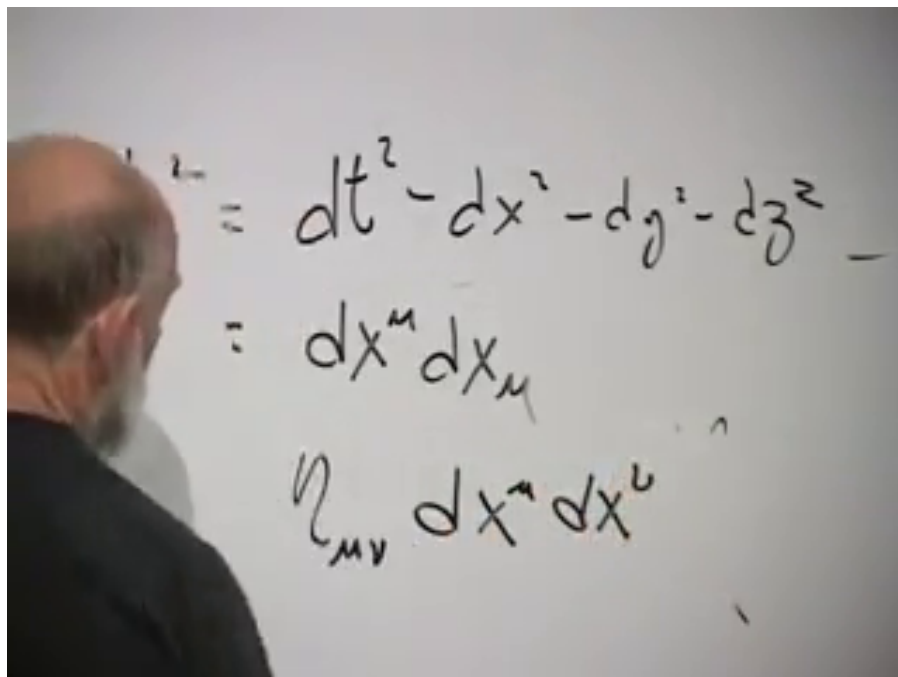


Figure 1.3: The Minkowski interval written successively in component, contracted-index, and metric-tensor notation.

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The simplicity of (1.6) depends on the coordinates. Even a flat plane can acquire unfamiliar coefficients if we label it unwisely. Let x be measured in centimeters while y is measured in meters. For a physical displacement,

$$ds^2 = dx^2 + \left(\frac{dy}{100}\right)^2 = dx^2 + 10^{-4}dy^2. \quad (1.7)$$

The factor 10^{-4} says nothing about curvature. It merely translates between units.

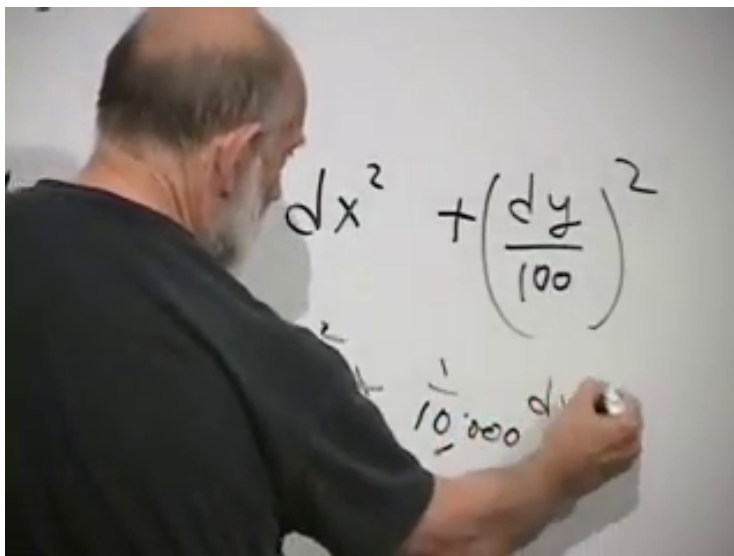


Figure 1.4: A flat plane described with centimeters on one axis and meters on the other. The unequal metric coefficients encode units, not curvature.

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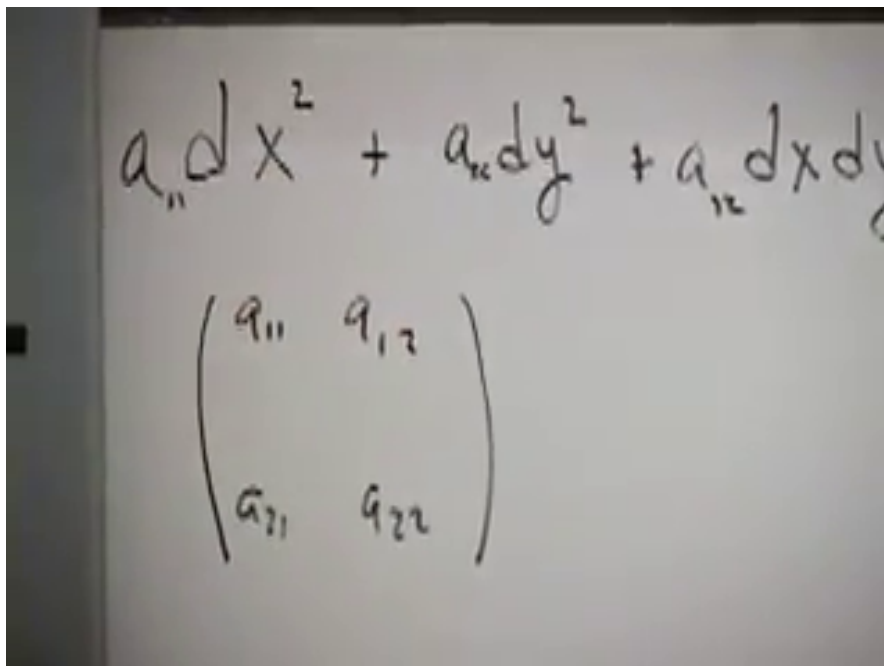
Oblique coordinates add a mixed term. In two dimensions the most general local quadratic form can be written

$$ds^2 = a_{11} dx^2 + a_{12} dx dy + a_{21} dy dx + a_{22} dy^2, \quad a_{12} = a_{21}. \quad (1.8)$$

Because ordinary differentials commute, $dx dy = dy dx$, only the symmetric part of the coefficient array contributes. We collect the coefficients into the matrix

$$g = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}. \quad (1.9)$$

This matrix is the *metric*. It is the local rule that converts coordinate differences into physical intervals.



The image shows a whiteboard with handwritten mathematical expressions. The top line is a quadratic differential form: $a_{11}dx^2 + a_{22}dy^2 + a_{12}dxdy$. Below it is a symmetric 2×2 coefficient matrix written as $\begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}$.

Figure 1.5: The quadratic differential form and its symmetric 2×2 coefficient matrix at the moment the matrix is identified as the metric.

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Uniform rectangular or oblique grids give constant coefficients. If the coordinate lines curve, bunch together, or spread apart, those coefficients become functions of position even when the underlying surface remains flat. The general spacetime notation is

$$ds^2 = g_{\mu\nu}(x) dx^\mu dx^\nu. \quad (1.10)$$

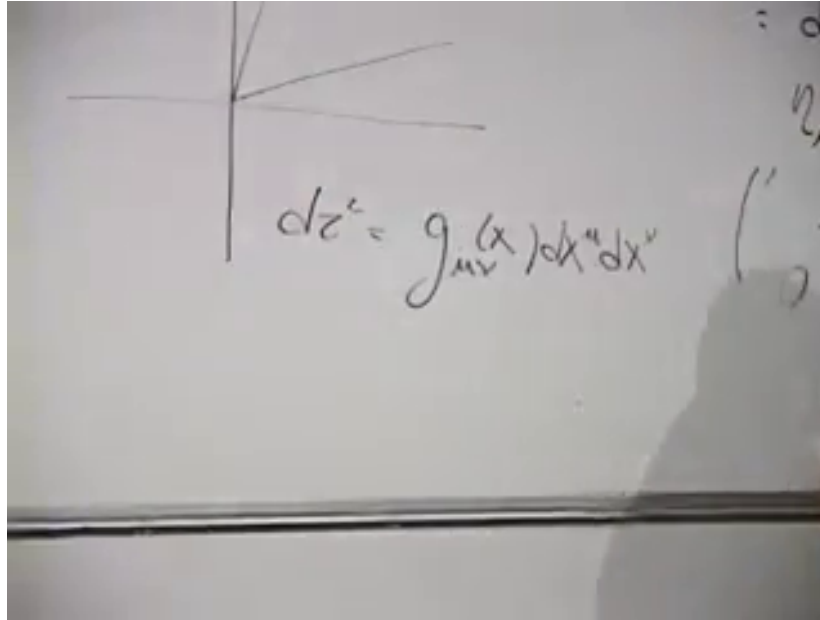


Figure 1.6: The general position-dependent metric after the coordinate grid is allowed to vary from point to point.

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Knowing $g_{\mu\nu}$ in a specified coordinate system determines local distances, times, and angles and thereby determines the geometry. The converse is not unique: one geometry has infinitely many coordinate descriptions, hence infinitely many arrays of coefficient functions related by coordinate transformations.

1.3 Coordinate complexity and intrinsic curvature

We must now distinguish a complicated metric from a curved space. Roll a sheet of paper into a cylinder without stretching it. Its appearance in the room changes, but all distances measured within the sheet remain the same. A creature confined to the paper cannot detect the bending by any intrinsic measurement. That is extrinsic bending, not the curvature meant here.

A hemisphere is different. It cannot be pressed flat without stretching, tearing, or compressing it. A flat map of Earth must distort either distance, area, angle, or some combination of them. This obstruction is intrinsic and can be detected without looking into a surrounding dimension.

Polar coordinates give the indispensable flat counterexample. On a plane, an infinitesimal radial displacement has length dr , while an angular displacement $d\theta$ at radius r has length $r d\theta$. Thus

$$ds^2 = dr^2 + r^2 d\theta^2, \quad g_{\text{polar}} = \begin{pmatrix} 1 & 0 \\ 0 & r^2 \end{pmatrix}. \quad (1.11)$$

The coefficient r^2 varies across the plane, yet the plane is flat. The change of variables

$$x = r \cos \theta, \quad y = r \sin \theta \quad (1.12)$$

returns the constant metric $ds^2 = dx^2 + dy^2$.

$$dr^2 + r^2 d\theta^2 \rightarrow dx^2 + dy^2$$

$$\begin{pmatrix} 1 & 0 \\ 0 & r^2 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

Figure 1.7: The polar metric and its return to the Cartesian metric. The position-dependent matrix on the left still describes a flat plane.

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Question from the audience. Should the angular distance in polar coordinates contain a sine?

Response. Not on the plane. There the arc length is $r d\theta$. A sine appears on the sphere, where a circle of latitude has radius $R \sin \theta$.

Take a sphere of radius R . Let θ be the polar angle measured from the north pole and ϕ the azimuthal angle. A meridional step has length $R d\theta$. At polar angle θ , the circle of latitude has radius $R \sin \theta$, so an azimuthal step has length $R \sin \theta d\phi$. Local Pythagoras gives

$$ds^2 = R^2 (d\theta^2 + \sin^2 \theta d\phi^2), \quad (1.13)$$

$$g_{\text{sphere}} = \begin{pmatrix} R^2 & 0 \\ 0 & R^2 \sin^2 \theta \end{pmatrix}. \quad (1.14)$$

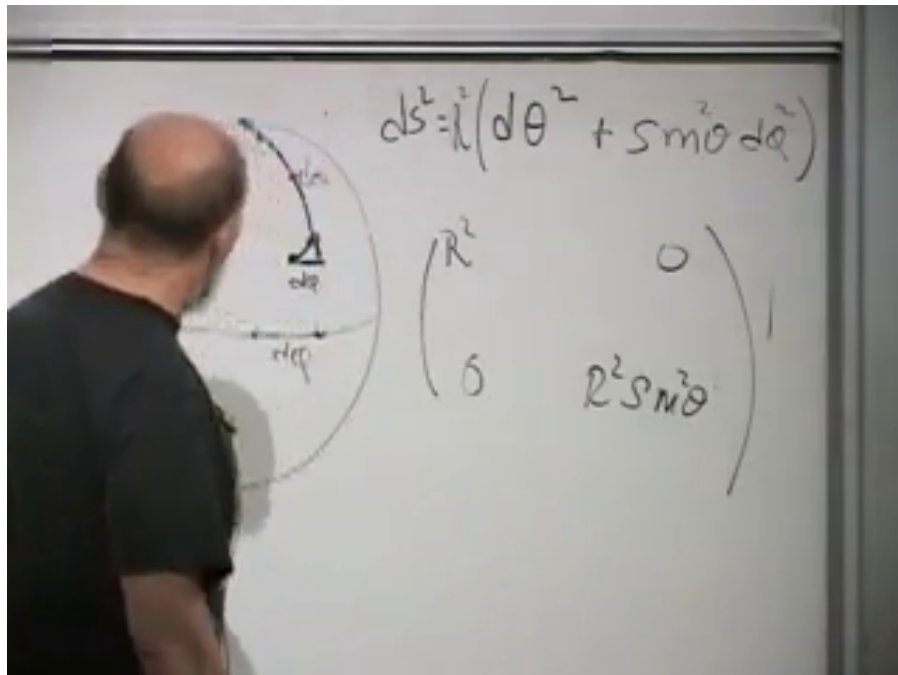


Figure 1.8: The sphere, its angular line element, and the corresponding metric matrix. The $\sin^2 \theta$ factor records the shrinking circles of latitude.

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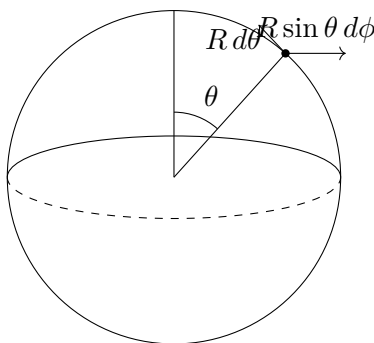


Figure 1.9: Clean reconstruction of the local differential legs used to obtain (1.13).

No global change of coordinates turns (1.13) into a constant Euclidean matrix. A two-dimensional observer can discover this intrinsically: a sufficiently large triangle made from geodesics has an angle sum greater than 180° . We have therefore reached the useful criterion. Variable coefficients do not by themselves imply curvature. Curvature is the obstruction to finding coordinates that make the metric constant throughout the region under consideration.

1.4 From curved spacetime to cosmology

General relativity permits spacetime itself to possess this intrinsic curvature. A position-dependent $g_{\mu\nu}$ may still be an artifact of coordinates, as in polar coordinates, but in a genuinely curved

spacetime no coordinate transformation removes the variation globally.

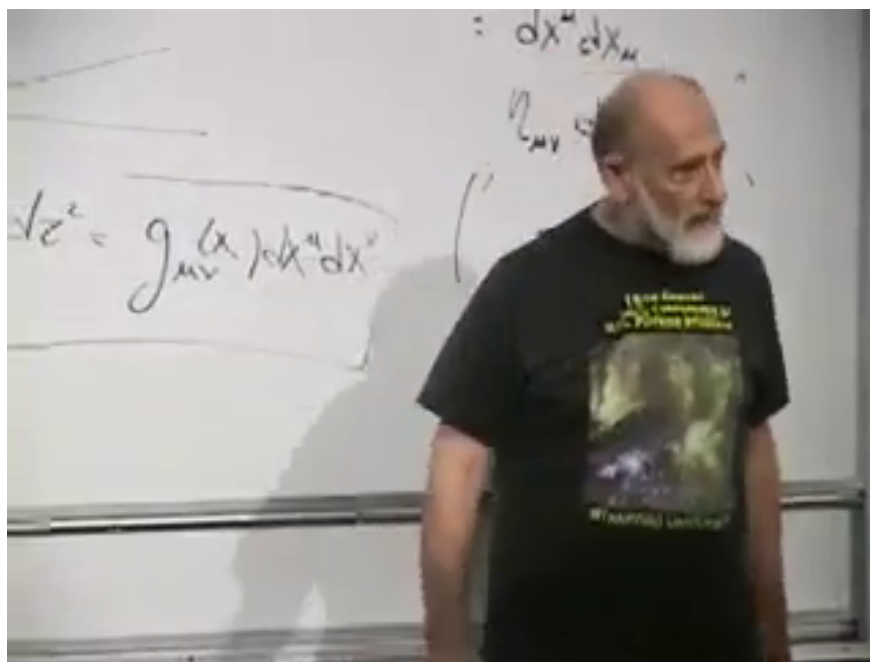


Figure 1.10: The transition from special coordinate examples to the general spacetime line element $ds^2 = g_{\mu\nu}(x)dx^\mu dx^\nu$.

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We will not attempt the full machinery of Einstein's equations. Instead we restrict attention to geometries appropriate to cosmology: space is taken to be homogeneous on sufficiently large scales, while the geometry may change with time. Homogeneous means that no location is distinguished from another. It does not mean flat: every point on a sphere is equivalent, yet the sphere is curved. It does not mean featureless on every scale either.

Question from the audience. How can the universe be homogeneous when galaxies form clusters, sheets, and filaments?

Response. The claim concerns coarse-grained scales. Stars, galaxies, and clusters are strongly inhomogeneous. When regions hundreds of millions of light-years across are compared, those structures average toward a uniform distribution. The cosmological principle is an empirical working hypothesis, not a logical necessity; sufficiently large systematic differences, including directional differences in the cosmic background, could disprove it.

The second large-scale assumption in this lecture is spatial flatness. At a fixed cosmic time, sufficiently large triangles appear Euclidean: their angles add to 180° . This is a statement about the geometry of three-dimensional space, not yet about the curvature of four-dimensional spacetime.

Question from the audience. What does simultaneity mean for points separated by cosmological distances, since their light reaches us at different times?

Response. We cannot place rods between distant galaxies or inspect them instantaneously. We receive signals later and reconstruct the geometry of the emitting events within a spacetime model. The procedure is indirect, but so are many geometric measurements in astronomy. Flatness means that the reconstructed large triangles show no detectable angular excess or deficit on the surveyed scale.

1.5 Comoving coordinates and the scale factor

Imagine a Cartesian grid painted onto the large-scale flow of galaxies. Each galaxy keeps its coordinate label as the grid expands. The label is not a distance in meters; it is an address. A time-dependent conversion factor turns coordinate separations into physical separations.

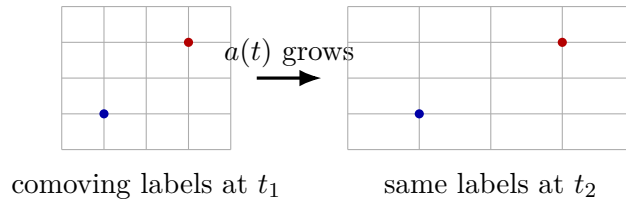


Figure 1.11: The comoving addresses stay fixed while the physical spacing of the coordinate grid grows.

Spatial homogeneity requires the same conversion factor everywhere at a given time. Spatial flatness allows the spatial part of the metric to be Cartesian. The resulting line element is

$$d\tau^2 = dt^2 - a^2(t) (dx^2 + dy^2 + dz^2). \quad (1.15)$$

This is the spatially flat homogeneous cosmological metric in comoving coordinates. We may choose x, y, z dimensionless and give $a(t)$ units of length. Another convention can put the dimensions into the coordinates and make a dimensionless; only the product matters.

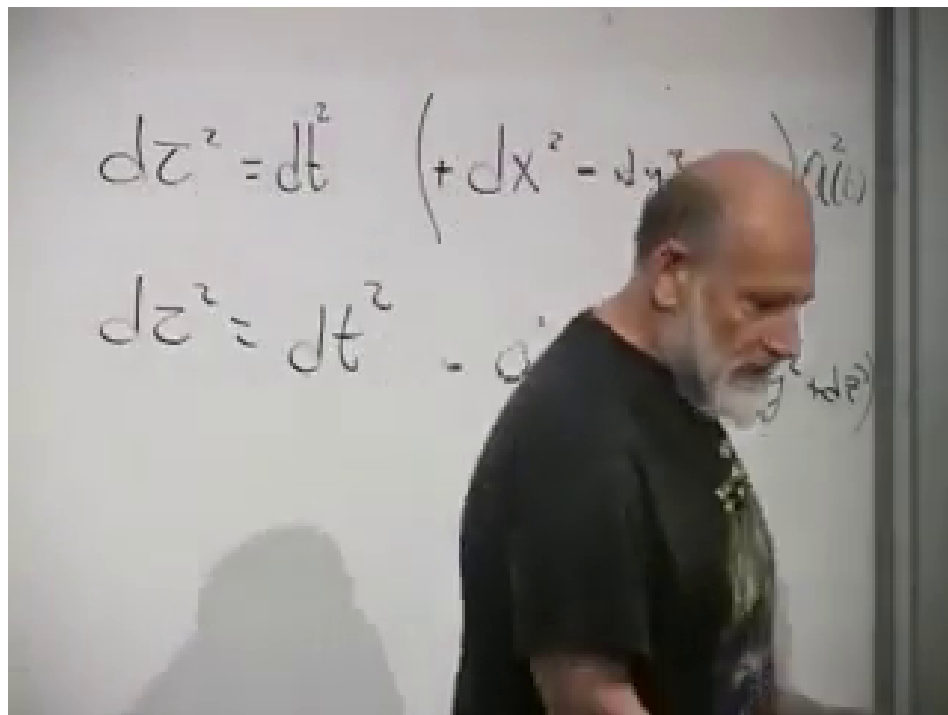


Figure 1.12: The spatially flat cosmological metric as completed on the board, with the common factor $a^2(t)$ multiplying all three spatial directions.

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If a were constant, we could absorb it by rescaling x, y, z , and (1.15) would be ordinary Minkowski spacetime in different units. Its time dependence is the nontrivial ingredient.

Question from the audience. If the coordinate grid expands, why do meter sticks, atoms, and galaxies not expand with it?

Response. The metric describes the average cosmic flow, not a force that compels every bound system to follow it. Electromagnetic forces hold a meter stick and an atom together; self-gravity holds a galaxy together. Those binding effects dominate the exceedingly weak background expansion on small scales. Nearby galaxies can also depart from the Hubble flow: Andromeda is pulled toward the Milky Way. Detailed studies of galactic binding show that visible matter alone is insufficient, one of the arguments for dark matter.

Question from the audience. Why is the scale factor not itself a force that pushes galaxies apart?

Response. It has the dimensions of length in the present convention, not force. Think of a cannonball launched upward. Once launched, its outward motion is inertial; gravity changes that motion by reducing its speed or pulling it back. Likewise, expansion is a state of motion. The dynamical question is how gravity and other sources change $\dot{a}$ and $\ddot{a}$, not what force is required to keep a moving.

The old expectation was that attractive gravity should decelerate the expansion. By the time of this lecture, astronomical evidence had shown that the recent expansion is instead accelerating, which means that the complete dynamics contains more than ordinary attractive matter. We postpone that issue until the geometrical setup is secure.

1.6 The Hubble law

Take two comoving galaxies whose fixed coordinate separation is Δx . At a common cosmic time their physical distance is

$$D(t) = a(t) \Delta x. \quad (1.16)$$

Differentiate while holding the comoving labels fixed:

$$\begin{aligned} V(t) &= \dot{D}(t) = \dot{a}(t) \Delta x \\ &= \frac{\dot{a}(t)}{a(t)} D(t). \end{aligned} \quad (1.17)$$

Define

$$H(t) = \frac{\dot{a}(t)}{a(t)}. \quad (1.18)$$

Then

$$V(t) = H(t) D(t). \quad (1.19)$$

At one cosmic time the same $H(t)$ applies everywhere; otherwise space would not be homogeneous. The recession speed is proportional to distance because every interval in the comoving grid is stretched by the same fractional rate.

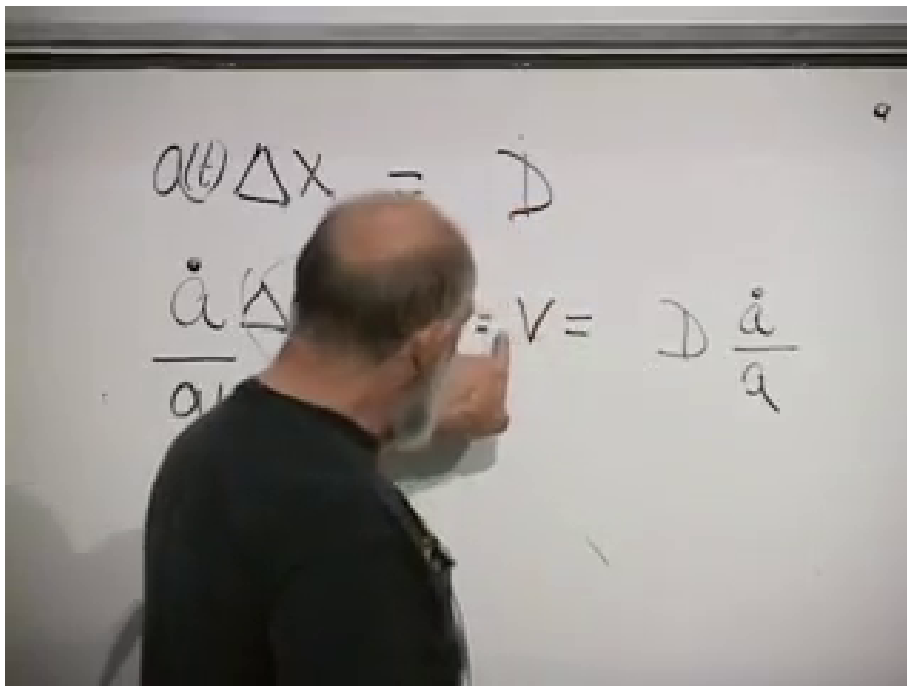


Figure 1.13: Physical distance $D = a\Delta x$ and its time derivative on the way to $V = (\dot{a}/a)D$.

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Question from the audience. For a distant galaxy, should the Hubble parameter be evaluated now or when the observed light was emitted?

Response. At the epoch being described. Light from a distant galaxy reports an earlier value of the expansion. A serious analysis must account for the time dependence of H ; inserting today's value everywhere is only a local approximation.

Question from the audience. Where do the units reside in $D = a\Delta x$?

Response. A convenient choice is to treat the comoving label Δx as dimensionless and let a carry length. The opposite bookkeeping is possible. Either way, $H = \dot{a}/a$ has dimensions of inverse time.

1.7 Light in the expanding coordinates

Light follows a null path. Along such a path the proper-time interval vanishes:

$$d\tau^2 = 0. \quad (1.20)$$

Question from the audience. Why is the interval zero for light?

Response. That is the defining geometry of a light ray in relativity. Every local freely falling observer measures light to move at $c = 1$, so $dt^2 = d\ell^2$ locally and the spacetime interval vanishes. It does not mean that no coordinate time passes for the observer receiving the light.

For a ray moving only in the x -direction,

$$0 = dt^2 - a^2(t) dx^2. \quad (1.21)$$

Choosing the ray that moves toward increasing x ,

$$dt = a(t) dx, \quad dx = \frac{dt}{a(t)}, \quad \frac{dx}{dt} = \frac{1}{a(t)}. \quad (1.22)$$

The image shows a whiteboard with handwritten mathematical equations. The top equation is $d\tau^2 = dt^2 - a^2(t) dx^2$. Below this, there is a rectangular box containing the equation $\frac{dt}{a(t)} = dx$.

Figure 1.14: The null condition in the expanding metric and the resulting comoving light-ray equation $dx = dt/a(t)$.

Lecture frame: 01:23:45.

The coordinate speed dx/dt decreases as the scale factor grows. This does not mean that light locally slows down. One unit of comoving coordinate distance represents an ever larger physical distance, so the numerical coordinate speed changes while every local measurement still gives $c = 1$.

1.8 Why the time dependence matters

The central geometrical question is now unavoidable: how does $a(t)$ vary? The analogy is a projectile whose position is known at one instant. Its future trajectory requires an equation of motion. In cosmology, gravitational dynamics will supply the equation for $a(t)$.

Question from the audience. Why privilege the question whether $a(t)$ changes over the equally legitimate question whether, for example, the electron charge changes with time?

Response. There is no logical hierarchy. A measured change in the electron charge would be enormously important. We focus on $a(t)$ because observation already tells us that cosmic separations change, whereas the electron charge appears constant. Geometrically, a constant a could be removed by a change of spatial units and would leave flat spacetime; a time-dependent a cannot be removed that way.

Before measuring distant recession, one needs this geometrical framework to say what is being measured. At the crudest level, a source of known luminosity appears dimmer with distance because its light spreads over a larger area, while the displacement of known atomic spectral lines gives a redshift and hence a recession indicator. Hubble combined distance estimates with redshifts to uncover the proportionality (1.19), although the early numerical value of the coefficient was substantially wrong. Modern distance ladders and relativistic light-propagation calculations are much more elaborate, but the conceptual pair remains distance plus redshift.

1.9 The Hubble time and the age estimate

There is a simple estimate hidden in the Hubble law. Freeze a galaxy's present recession speed and run the motion backward uniformly. The time required to close its present distance would be

$$t = \frac{D}{V}. \quad (1.23)$$

Using $V = HD$,

$$t = \frac{D}{HD} = \frac{1}{H}. \quad (1.24)$$

The distance cancels, so every pair of comoving galaxies gives the same rough time. Susskind catches and corrects an easy verbal error here. The assumption is not that H remains constant. It is that the particular recession velocity is extrapolated backward as though it were constant.

$$\frac{1}{H} = \frac{D}{HD} = \frac{D}{V} = t$$

Figure 1.15: The Hubble-time estimate $H^{-1} = D/(HD) = D/V$, with distance canceling from the backward extrapolation.

Lecture frame: 01:33:45.

The inverse Hubble parameter is therefore a time scale, not automatically the exact age of the universe. In a purely decelerating, matter-dominated picture, reverse the present velocity and let gravity pull the galaxies together. They meet sooner than they would at uniform velocity, so the present H^{-1} overestimates the age. The lecture quotes a refined age of about 13.5 billion years, close to the Hubble time but not identical to it.

Question from the audience. If there is no Hubble expansion inside our galaxy, what does it mean to run all galaxies backward until they collapse?

Response. The homogeneous description eventually ceases to resolve the real history of bound structures. Forward in time, small fluctuations grow gravitationally into galaxies and clusters. A literal reversed movie would require every microscopic motion to reverse and entropy to decrease; simply reversing the average Hubble flow does not undo structure formation. The backward extrapolation is a coarse cosmological estimate, not a film of every star retracing its path.

Question from the audience. Can an experiment distinguish an expanding intergalactic distance from a shrinking ruler?

Response. Only dimensionless ratios are observable. What is measured is that the distance between galaxies divided by an atomic or laboratory length standard increases. We conventionally hold the parameters of atomic and molecular physics fixed and say that the universe expands. One could choose the intergalactic separation as the standard and describe atoms, protons, and meter sticks as shrinking. The physics of the ratio would be unchanged.

That convention is natural because rods are made from atoms and their size is controlled by parameters that appear constant. A meter is ultimately a large number of microscopic lengths. Holding those microscopic standards fixed gives the familiar statement that $a(t)$ grows.

Question from the audience. Is (1.15) the only metric compatible with spatial homogeneity and flatness?

Response. Only up to coordinate transformations. The precise claim is that, under those assumptions and a suitable choice of cosmic time and comoving spatial coordinates, the metric can be brought to this form. Another coordinate system may make the same spacetime look quite different.

Question from the audience. When the backward separation reaches zero, the Hubble law also gives zero velocity. Does that prevent the galaxies from meeting?

Response. No. The H inferred today is not held fixed all the way back. In ordinary expanding solutions $H(t)$ was much larger at earlier times. The estimate H_0^{-1} is made by extending today's velocity uniformly only to obtain a scale; the real $a(t)$, $H(t)$, and velocities must be found from dynamics.

1.10 Newton's shell theorem and cosmological dynamics

To discover the dynamics, smear the galaxies into a homogeneous medium. Choose one galaxy as the origin and another with comoving label x . There is no physical privilege in this choice; homogeneity requires the final equation to be the same around every galaxy. The physical radius of the second galaxy is

$$R(t) = a(t)x. \quad (1.25)$$

At first, finding its gravitational acceleration seems to require summing the attraction of every galaxy in an enormous universe. Newton's shell theorem collapses the problem:

Proposition 1.1 (Newton's shell theorem). *For an inverse-square force and a spherically symmetric mass distribution, matter outside a sphere through the test point contributes zero net force. The matter inside the sphere acts on the test point as though all enclosed mass were concentrated at the center.*

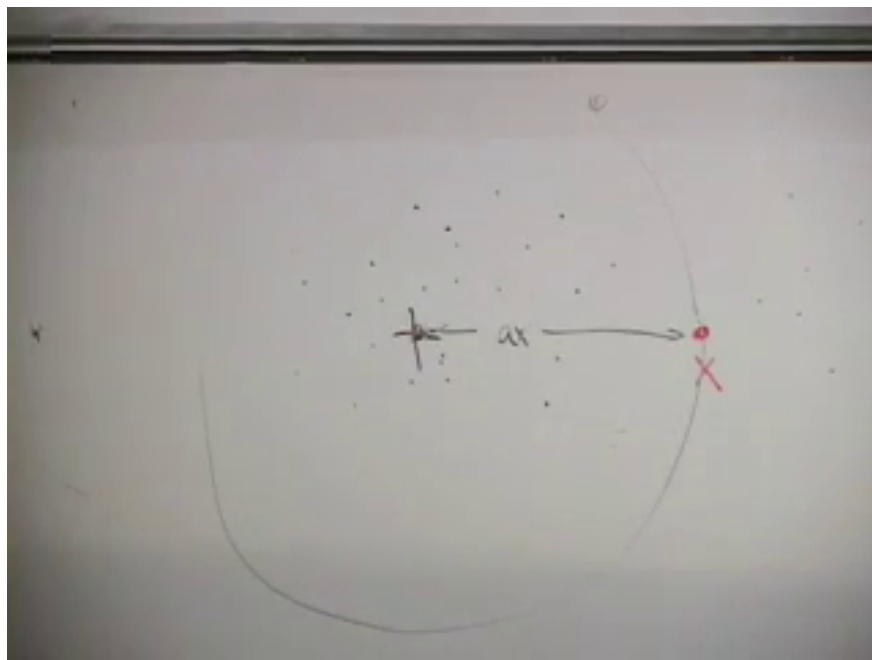


Figure 1.16: The homogeneous galaxy distribution, a test galaxy at comoving label x , and the sphere whose enclosed mass alone controls its radial motion.

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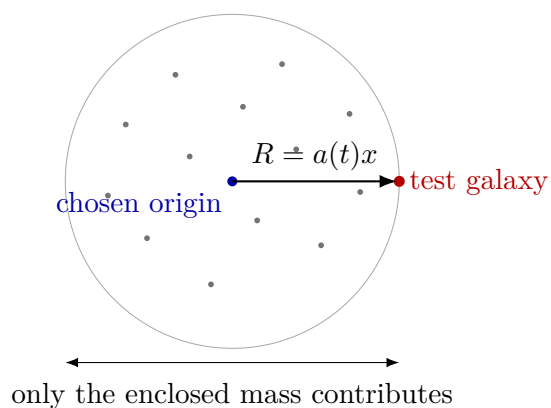


Figure 1.17: Clean reconstruction of the shell-theorem setup. The origin is a convenient choice, not a center of the universe.

The cosmological many-body problem has therefore become a familiar one-body problem: a projectile moving radially under the attraction of the mass enclosed inside its present sphere. Newton's equation will determine how $R = ax$, and hence $a(t)$, accelerates. The lecture deliberately stops before carrying out that calculation. The next meeting will derive the simplest matter-dominated cosmology rather than smuggling its answer into the geometrical setup.

1.11 Late questions: light, the Big Bang, and curvature

The formal lecture has ended, but the final questions expose several misleading pictures and are part of the physics rather than an appendix.

Question from the audience. If light from very distant objects was emitted when the universe was much smaller, why does it now appear to come from so far away? And if the early universe was compact, why is not all of it observable?

Response. One must solve the null-ray equation in the expanding coordinates. When $a(t)$ is very small, $dx/dt = 1/a(t)$ is very large: light crosses a large comoving interval while still moving locally at $c = 1$. As $a(t)$ grows, its comoving coordinate speed decreases. The precise emission point and the size of the observable region depend on the entire history of $a(t)$, so a numerical answer is postponed until a specific cosmology has been solved.

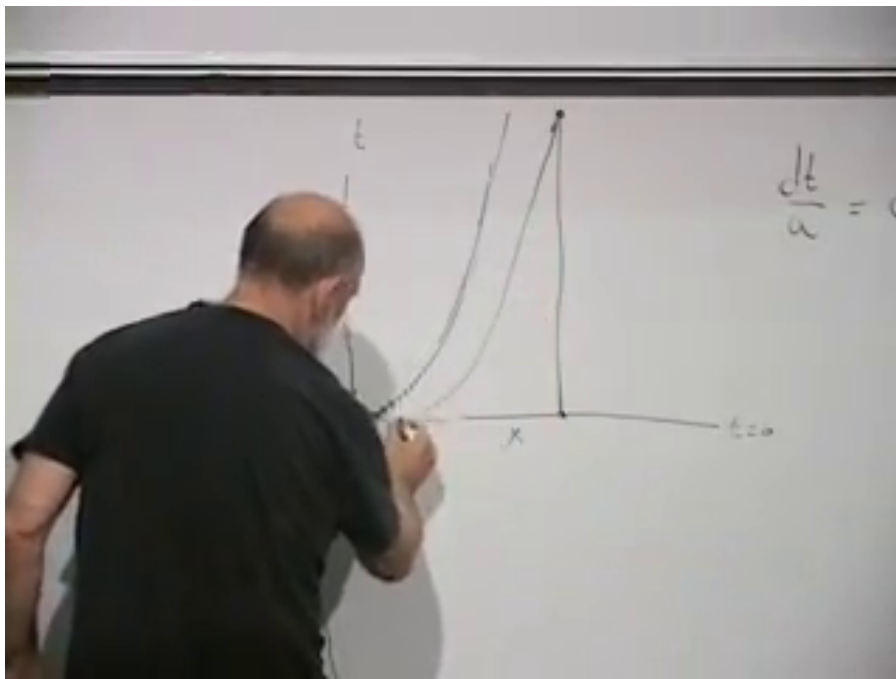


Figure 1.18: A light ray in the (x, t) plane: nearly horizontal in comoving coordinates when $a(t)$ is small, then progressively steeper as the scale factor grows.

Lecture frame: 01:56:20.

Integrating (1.22) makes the deferred calculation explicit:

$$\Delta x = \int_{t_e}^{t_0} \frac{dt}{a(t)}. \quad (1.26)$$

Whether the integral reaches arbitrarily large comoving distances as t_e approaches the beginning depends on the early-time form of $a(t)$.¹

¹Editorial clarification: Equation (1.26) is the integrated form of the board equation $dx = dt/a(t)$. It makes explicit the calculation that the lecture assigns to the next meeting without choosing a particular scale factor.

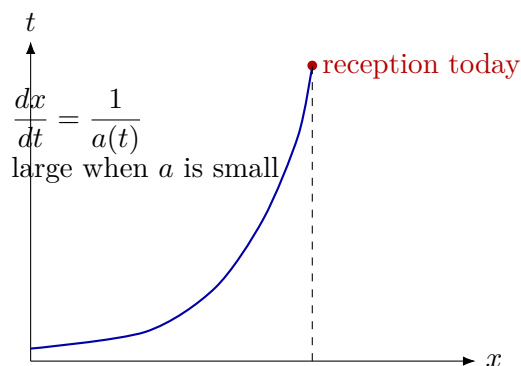


Figure 1.19: Clean version of the light-ray construction in comoving coordinates. The changing slope is a coordinate effect, not a changing local speed of light.

Question from the audience. Did the Big Bang eject matter from one point so that it should now lie on an expanding shell or ring?

Response. No. The model begins with a homogeneous, extremely dense space. Every large-scale separation grows; matter is not launched through pre-existing empty space from a preferred center. A balloon covered uniformly with dots is a useful two-dimensional analogy: as it expands, all dot separations grow while the surface remains uniformly populated. The space may even be infinite at every positive time; expansion does not imply that it began as a finite ball inside something else.

Question from the audience. Was gravity itself weaker or stronger during the expansion?

Response. The discussion keeps Newton's constant G fixed. The force nevertheless changes because the separations and enclosed densities change. Earlier, the same comoving galaxy was physically closer and the homogeneous matter was denser, so the gravitational deceleration was larger. This is precisely what the shell-theorem calculation is meant to quantify.

Finally, spatial curvature must not be confused with expansion. A small patch of Earth appears flat because its size is tiny compared with Earth's radius. By measuring sufficiently large triangles, one can either detect an angular excess or place a lower bound on the radius of curvature. Astronomy does the analogous experiment with enormous triangles. In the observational language used in the 2006 lecture, no curvature was detected across the visible region, so any curvature radius had to be substantially larger than that region. This does not prove an exactly flat universe; it bounds how curved the accessible part can be.

For a sphere, intrinsic curvature scales as R^{-2} . A growing sphere therefore becomes locally flatter. Likewise, if a homogeneous universe has nonzero spatial curvature, expansion increases its curvature radius and reduces the curvature visible in any fixed-size local patch. ²

²Editorial clarification: The R^{-2} scaling is the standard quantitative form of the lecture's statement that the numerical curvature decreases as a sphere grows.

Question from the audience. If space is curved, must one component of the universe be contracting while another expands?

Response. No. Positive, zero, or negative spatial curvature describes the geometry of each cosmic-time slice. Expansion or contraction is controlled by the evolution of $a(t)$. A curved universe can expand; a flat universe can contract. Expansion may later reverse, just as an upward-moving projectile may fall back, but simultaneous contraction is not hidden inside curvature. The dynamical issue is whether the outward motion escapes the pull of the enclosed mass, falls back, or lies on the critical boundary.

That escape-velocity comparison is the last board-level idea of the lecture. Geometry has supplied the metric, the Hubble law, and the light-ray equation. Newton's theorem has reduced the dynamics to the mass inside one sphere. The next lecture begins exactly where this one should end: with the equation that determines $a(t)$, and with the question of whether the expansion escapes, recollapses, or sits on the knife edge.